BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: PR–XII (A-I-A) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** PR–XII (A-I-A) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The logarithmic form of b^y=x is = _____.

(A) log_b x=y
(B) log_b y=x
(C) log^x_y=b
(D) log^y_10=x

Q2. If f(x) = $(x^2-3)/(x-1)$, then f(1) = _____.

(A) −1
(B) 1
(C) 0
(D) Undefined

Q3. If f(x) = sin(x) then f'(x) is = _____.

(A) −cos(x)
(B) sin x
(C) −sin x
(D) cos x

Q4. lim(x→a) (a^x −1)/x, a>0 = _____.

(A) lna
(B) lnx
(C) 0
(D) Does not exist

Q5. $d/dx(\cos^{-1}x)$ = _____.

(A) cosu
(B) −cosu
(C) −sinu du/dx
(D) −cosu du/dx

Q6. $d/dx(\cos^{-1}x)$ = _____.

(A) $1/\sqrt{(1-x^2)}$ (B) $-1/\sqrt{(1-x^2)}$

(C) $1/\sqrt{(x^2-1)}$ (D) $-1/\sqrt{(x^2-1)}$

Q7. The critical value of $f(x) = x^2/(x-2)$ are = _____.

(A) 0,2 (B) 0,2,4

(C) 0,4 (D) 0,1,2

Q8. ∫e^x dx = _____.

(A) xe^x+c (B) e^x+c

(C) xe^x−1+c (D) x+c

Q9. The function f(x) is concave upward on (a,b) if _____.

(A) f''(x)≤0 (B) f''(x)≥0

(C) f''(x)>0 (D) f''(x)<0

Q10. If for all a<x<b and f'(x)<0, then f(x) is strictly _____.

(A) Increasing (B) Decreasing

(C) Relative maximum (D) Relative minimum

Q11. The distance between (0,1) and (−1,3) is = _____.

(A) 0 (B) 1

(C) 3 (D) √5

Q12. The slope of line y−x+c is = _____.

(A) 1 (B) 2

(C) 3 (D) 2.33

Q13. Which of the given is homogenous equation _____.

(A) $3x^2y-4xy+5y^2=0$ (B) $4xy^2+5y^2=0$

(C) $x^2y-4xy^2=0$ (D) $3x^2y-4xy^2+y^2=0$

Q14. The coordinate of center in $(x-1)^2+(y-2)^2=r^2$, are _____.

(A) (h,k) (B) (1,2)

(C) (0,0) (D) (−h,−k)

Q15. The shorter axis of an ellipse is called the _____ axis.

(A) Major (B) Minor

(C) Transverse (D) Conjugate

Q16. The eccentricity e, of the conic is parabola if _____.

(A) $e<1$ (B) $e>1$

(C) $e\leq1$ (D) $e=1$

Q17. If $f(x,y)=\sqrt{(x^2-y^2+2)}$, then $f(1,-1)$ is _____.

(A) 1 (B) 2

(C) 3 (D) 4

Q18. The solution of $dy/dx=\cos x$ at $y(0)=1$, is _____.

(A) $\sin x+1$ (B) $-\sin x+2$

(C) $-\sin x+1$ (D) $\sin x+2$

Q19. The latus rectum in $y2 = -4ax$, is _____.

(A) $x=a$ (B) $x=-a$

(C) $y=a$ (D) $y=-a$

Q20. The degree and order of $d^2y/dx^2 - dy/dx = 0$ is _____.

(A) (1,2) (B) (2,1)

(C) (2,2) (D) (1,1)

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FIVE marks.

1) Sketch the graph of $x^2/9 + y^2/16 = 1$.

2) Find f'(x), if $f(x) = x\sqrt{(2x^2+12)}$

3) If $f(x) = 4\cos(3x+8)$, find 5th derivative.

4) Find the velocity and acceleration of the particle at a time t=0, whose position vector is $\bar{r} = 5t\hat{i} + 3t\hat{j} - \cos(t)\hat{k}$,

5) Evaluate the integral $\int(3x^2+8)(x^3+8x)^7\,dx$

6) Find the angle form the line $7x+3y-2=0$ to the line $5x-2y+9=0$.

7) For what value of n, the line $y = -x + c$, touch the circle $x^2 + y^2 = 16$.

8) Find the equation of tangent and normal to parabola $y^2=4x$, at $(2,-4)$

9) For what value of c the line $y=x+c$, will touch the hyperbola $x^2/25 - y^2/16 = 1$

10) Solve the initial value problem $dy/dx = 3x^2$, $y(0)=1$

11) Find f_x, f_y at point (π/3, 0) in f(x) = $x^2 \sin(x^3 + 3y)$

12) Use bisection method to solve f(x) = e^x − x − 2, [1,2], compute five iterations.

13) Find the area of the triangular region whose vertices are A(−3,6), B(3,2), C(6,0).

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 16

> Note: Note: Attempt any THREE questions. All questions carries equal marks.

Q. Attempt any THREE questions. All questions carries equal marks.

Q3.

(a) Evaluate $\lim_{x\to 1} (\sqrt{x}-1)/(x-1)$

(b) Find an equation of tangent and normal to the curve $y = -x^2 + 9$ at (3,0), (−3,0)

Q4.

(a) Evaluate $\int (x+1)/(x^2+2x+2)\, dx$

(b) $\int_0^2 12x(x^2-4)^5\, dx$

Q5.

(a) Find the equation of parabola having focus at F(0,3), directrix = −3.

(b) Find the point of intersection of the line y=2x−2, and the ellipse $x^2/9 + y^2/4 = 1$.

Q6.

(a) Solve the differential equation dy/dx = 3 dy/dx − 2y

(b) Use Simpson's rule to find the solution of $\int_1^3 dx/x$, n=6